QUANTCORTEX:

AUDITING FINANCIAL ML BACKTESTS WITH EVALUATION CONTRACTS AND EXECUTABLE CHECKS

NeurIPS 2026 Workshop: TAE (Trust-AI-Eval): Can We Trust AI Evaluation?

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Motivation

A backtest turns portfolio decisions into holdings, cash flows and returns. Execution timing, cash income and trading costs can change how the strategies rank.

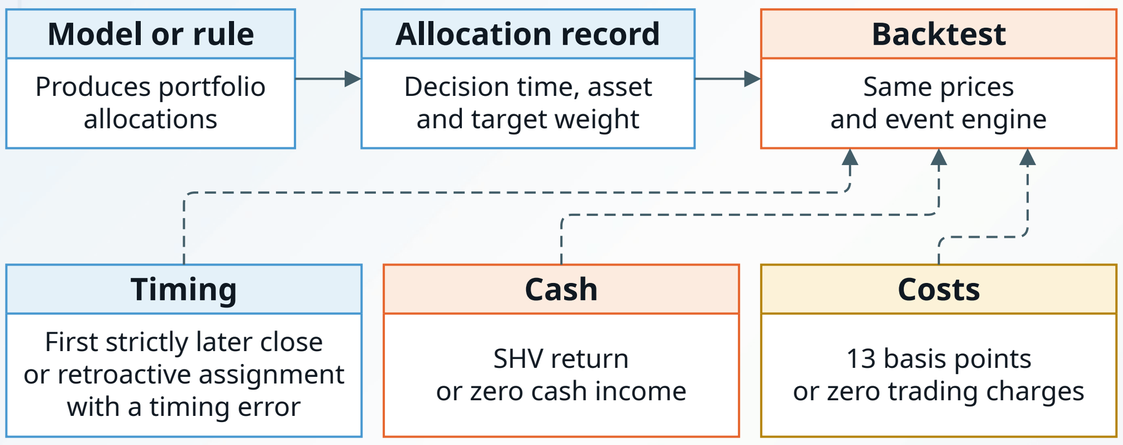
System design

Area	Baseline rule	Check
Allocations	Complete asset weights	Bounds and exposure
Timing	First strictly later close	Information and alignment
Cash / costs	SHV; purchases plus sales	Cash and ledger totals
Holdings	Units unchanged between trades	Portfolio value reconciles

Baseline charge: 13 basis points per dollar bought or sold.

Methods

Timing, cash and cost comparisons



Each comparison changes one setting above and reuses the same portfolio decisions.

Daily return decomposition

$$r_t^{\text{net}} - r_t^{\text{cash}} = \underbrace{(r_t^{\text{gross}} - m_t)}_{\text{allocation}} + \underbrace{(m_t - k_t)}_{\text{timing}} + \underbrace{(k_t - r_t^{\text{cash}})}_{\text{passive exposure}} + \underbrace{(r_t^{\text{net}} - r_t^{\text{gross}})}_{\text{costs}}.$$

m: basket and cash return at actual exposure; k: at full-sample mean exposure.
Daily arithmetic returns; timing describes historical association.

Uncertainty

Pointwise 95% intervals use 5,000 circular resamples of 21-session blocks.
The same dates apply to compared paths. Intervals are conditional on the original fits, decisions and seeds.

Study design

Historical weekly rotation: six ETFs with the original QQQ scoring defect.

Model comparison: three rules and gradient boosting regression trees (GBRT).

Nine sector ETFs and ten country ETFs, with monthly decisions.

Both studies cover 2018 to 2025. GBRT uses five seeds.

Rankings

	Baseline		Retroactive (timing error)		Zero trading cost	
Strategy	Sharpe ratio	Rank	Sharpe ratio	Rank	Sharpe ratio	Rank
United States sectors						
Time series momentum	0.494	3	0.526	2	0.525	4
Cross-sectional momentum	**0.596**	1	**0.572**	1	**0.630**	1
Short-term reversal	0.439	4	0.055	4	0.531	3
Gradient boosting	0.529	2	0.409	3	0.606	2
Country equities						
Time series momentum	-0.018	4	0.044	3	0.022	4
Cross-sectional momentum	**0.329**	1	**0.313**	1	0.364	2
Short-term reversal	0.197	3	-0.186	4	0.297	3
Gradient boosting	0.314	2	0.244	2	**0.413**	1

Sample Sharpe relative to SHV; GBRT averages five individual Sharpes. Bold: highest Sharpe.
Amber: rank differs from baseline. Retroactive assignment credits returns before the decision.

Return differences

Retroactive timing can raise or lower returns and move the first entry cost outside the sample.
GBRT averages five within-seed effects. Only the first three groups change one convention.

Backtest

Net growth trails SHV. In arithmetic means, allocation and costs outweigh the positive passive and timing contributions. Net excess and timing intervals include zero.

Conclusion

Accounting choices changed returns and sample rankings even when portfolio decisions stayed the same.
Report execution timing, cash treatment, trading charges and accounting checks with each performance claim.

These comparisons use historical data available during development.

github.com/magnaprog/quantcortex